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Eocene (51 Ma) end to northward translation of the Coast Plutonic Complex: paleomagnetism and K–Ar dating of the White Pass Dikes

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Abstract

The White Pass mafic dikes are located in the Coast Belt, inboard of the Coast Shear Zone–Denali Fault System, east of Skagway, Alaska, astride the Alaska–British Columbia border. The dikes intrude Eocene felsic plutons in the Coast Plutonic Complex. Three K–Ar analyses date the dikes at 51.9 ± 0.8 Ma (biotite), 50.1 ± 1.0 Ma and 45.0 ± 0.8 Ma (both whole rock). Structural analysis on these vertical dikes indicates that the study area has not been tilted significantly since dike emplacement. Conventional alternating field and thermal step-demagnetization and saturation-remanence paleomagnetic methods were used to analyse 407 specimens from 27 dike and five host rock sites. Three contact tests show that the specimens retain a primary thermoremanent magnetization (TRM) carried in pseudosingle to multidomain magnetite, although some magnetization in a few dikes is carried by pyrrhotite. The dual-polarity TRM is mostly reversed and the 24 accepted dike sites give a pole position of 245.7° E, 76.3° N ($d_p = 4.2°$, $d_m = 4.4°$), indicating a paleolatitude discordance of $8 \pm 4°$ south and a clockwise rotation of $40 \pm 9°$ with respect to the North American craton. These data, along with other paleomagnetic data on Eocene rocks, indicate that the allochthonous eastern Coast and Intermontane Belt terranes had halted their northward translation by about 51 Ma but continued to be rotated clockwise by an average of $24 \pm 10°$ thereafter, probably on underlying sole faults.
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1. Introduction

Much research effort using geologic correlations, paleomagnetism, fossil distributions, zircon sources, etc., has been spent over the past quarter century to determine the timing and amount of translation and rotation for the accreted terranes of the North American Cordillera (Monger, 1997). Nevertheless, a great deal has yet to be learned about the motion of the components of this collage in time and space. For any terrane or group of terranes it is important to know when displacement ended or, in paleomagnetic terms, when poles concordant with cratonic pole positions are first observed. At present there is good evidence that statistically significant northward motion ended for terranes underlying the Intermontane Belt of

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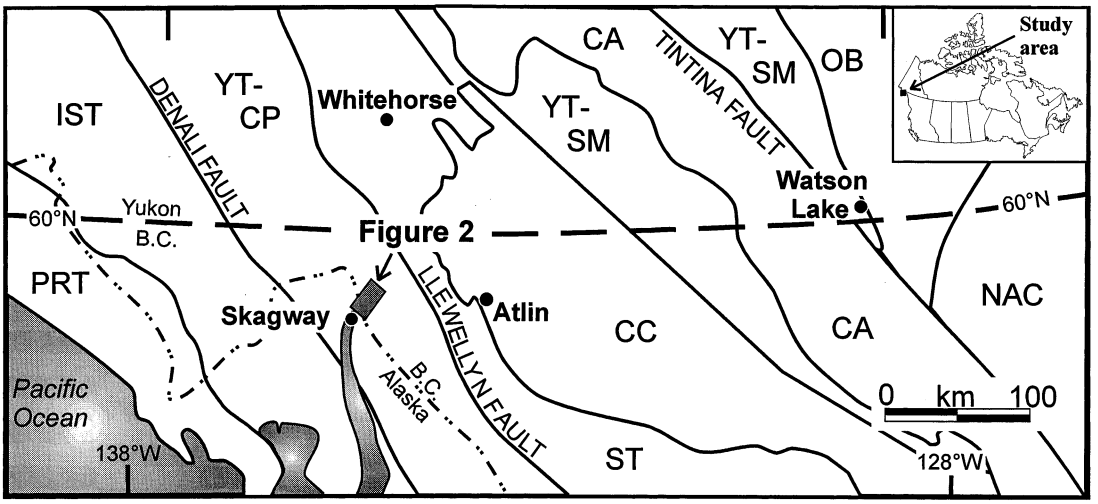


Fig. 1. Location map.

southern and central British Columbia (Fig. 1) by the Eocene (Fox and Beck, 1985; Bardoux and Irving, 1989; Symons and Wellings, 1989; Vandall and Palmer, 1990). There is no such constraint for terranes underlying the northern Intermontane Belt. Moving outboard, there are no well-defined concordant poles to establish the end of motion in the Coast Plutonic Complex (CPC), although some results are supportive (Symons, 1974, 1977). Further, some results from the Wrangellia terrane in the southern Alaska panhandle also suggest a Middle Eocene end to motion (Thrupp and Coe, 1986; Hillhouse and Coe, 1994). This study of the White Pass dikes was undertaken to examine well-exposed mafic dikes along the South Klondike Highway — the Skagway Traverse of Barker et al. (1986). These dikes intrude Eocene to older plutons in the CPC, and thus a result would help to constrain the end of the complex's accretionary motion.

2. Geology

The CPC, or Coast Batholith of many American workers (e.g. Brew et al., 1991), extends for 1500 km from Washington State, through British Columbia and into southeast Alaska and the Yukon. It formed as a coast-parallel Andean-type magmatic continental arc during the Cretaceous and early Tertiary. It is characterized by massive and foliated felsic to intermediate plutons that intruded into the western margin of the Intermontane terranes and eastern Insular components during orogenic collision. The Intermontane terranes, those underlying the Intermontane Belt, include the Stikine, Cache Creek and Quesnel terranes (Fig. 1), and the Insular terranes are dominantly Wrangellia and the Alexander Terrane. In the northern Cordillera, the Paleozoic and older metasedimentary rocks of the Yukon–Tanana Terrane are also considered part of the Intermontane terranes (Johnston et al., 1994) and host the CPC plutons in the study area (Brew et al., 1991; Monger, 1997).

In the study area, the western boundary of the CPC is marked by the Coast Shear Zone–Denali Fault Zone system (Fig. 1), which has been a locus of strike-slip motion throughout much of the Cretaceous and Tertiary (Hollister and Andronicos, 1997). The eastern boundary at the 60° N latitude of the study area is a modified intrusive contact that is now delineated by the Llewellyn Fault between the Yukon–Tanana and Stikine terranes.

Along the South Klondike Highway traverse the sampled mafic dikes intrude five younger plutons. The Log Cabin granite is the oldest pluton,

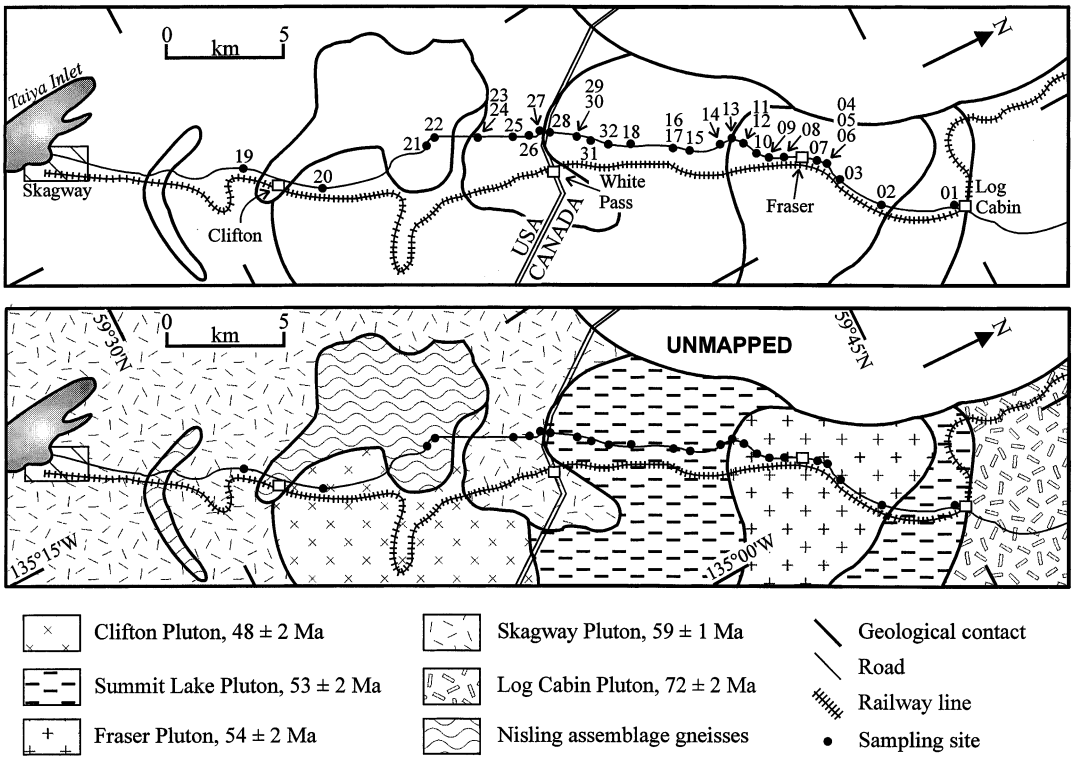


Fig. 2. Study area along the Klondike Highway through White Pass showing the geology modified from Johnston et al. (1993) with site locations.

yielding an age of 72 ± 2 Ma (Barker et al., 1986) (Fig. 2). Next oldest is the Skagway pluton composed of biotite granodiorite and biotite tonalite, which has been given a U–Pb zircon age of 58.6 ± 0.9 Ma (Gehrels et al., 1990); then the hornblende-biotite tonalite of the Fraser pluton has yielded a U–Pb zircon age of 54 ± 2 Ma (Barker et al., 1986); then the essentially coeval Summit Lake pluton is composed of coarse-grained biotite granodiorite to biotite granite and has a U–Pb zircon age of 53 ± 2 Ma (Barker et al., 1986); finally, the youngest is the Clifton pluton, also a biotite granodiorite to biotite granite with porphyritic phases, that has been dated at 48 ± 2 and 48.8 ± 1.0 Ma by U–Pb zircon techniques (Barker et al., 1986; Gehrels et al., 1991).

If the mafic dikes that cut these Eocene plutons were emplaced in one pulse, then they would have to postdate the Clifton granite at 48 ± 1 Ma. However, as noted later, there are good reasons to believe that the dikes were emplaced in multiple events. Thus the "average" age of the four plutons of 53.5 ± 3.5 Ma represents a reasonable upper limit age of the sampled dike swarm(s) for the purposes of selecting the paleomagnetic reference pole.

3. Sampling

Samples for analysis were collected at 27 sites in mafic dikes and at five sites in their granitic host rocks, the latter primarily for paleomagnetic contact tests. The sites are located in road cuts along a 35 km stretch of the South Klondike Highway through White Pass above the treeline, astride the Alaska–British Columbia border northeast of Skagway, Alaska (Fig. 2). Five to seven cores were drilled and oriented in situ by sun and/or magnetic compass at each site. The two

Table 1
Site mean remanence data

Site	Lithology[a]	Specimens[b]			Declination (deg)	Inclination (deg)	k[c]	α_{95} (deg)[d]	Note[e]
		e	r	o					
1	Dp	0	0	11					
2	Da	7	0	3	192.4	−75.0	61.6	7.8	1R
3	D	2	6	2	183.2	−74.6	44.7	8.4	1R
4	Dp	9	0	1	0.6	73.3	746.3	1.9	1N
	Gb	4	0	0	14.4	69.3	47.4	13.5	
	C	13	0	0	5.7	72.1	146.0	3.3	
5	Da	10	0	0	257.3	−54.4	50.4	6.9	2
	Gb	4	0	0	31.7	−54.2	5.2	61.0	
	C	14	0	0	270.5	−64.7	6.6	17.5	
6	Gt	0	0	12					
7	Dpa	0	0	14					3
8	Dp, l	9	0	3	345.2	72.2	39.8	8.3	1P
	Dp	0	10	2	233.0	−83.6	100.8	4.8	1R
9	Dp, l	9	0	4	32.3	75.7	48.3	7.5	1P
	Dp	2	8	3	219.9	−80.5	84.2	5.3	1R
10	Dp	3	7	0	161.2	−81.7	147.9	4.0	1R
11	Dp	0	11	1	139.6	−82.1	54.4	6.2	
	Gb	4	0	0	109.2	−77.7	216.7	6.3	
	C	4	11	1	128.6	−81.2	64.2	4.8	1M
12	Gt	9	2	0	283.0	80.9	44.9	6.9	1G
13	D	11	0	2	3.1	72.9	84.4	5.0	1N, 4
14	D	14	0	0	237.0	−79.0	119.8	3.7	1M
15	Dpa	5	7	2	220.8	−74.6	50.4	6.2	1R
16	Dp	12	2	0	264.9	−84.5	289.8	2.3	1R
	Gb	4	0	0	176.5	−71.4	23.7	19.3	
	C	16	2	0	244.0	−84.1	147.0	2.9	
17	Gg	6	0	7	26.1	71.2	22.2	14.5	1G
18	Dpa	11	0	4	207.4	−81.6	77.6	5.2	1R
19	Dp	11	0	1	145.9	−78.7	370.3	2.5	1R
20	Da	14	0	0	213.1	−75.7	78.4	4.5	1R
21	Dp	12	0	1	15.8	72.9	104.3	4.3	1N
22	Dp	14	1	0	158.1	−81.8	76.1	4.4	1R
23	Dpa	5	8	0	194.2	−85.0	87.1	4.5	
	Gb	1	2	1	164.3	−81.6	369.2	6.4	
	C	6	10	1	185.9	−84.5	98.9	3.7	1R
24	Gt	2	6	2	195.7	−81.1	25.4	11.2	1G
25	Dp	3	7	0	240.5	−82.7	87.2	5.2	1R
26	Dp	3	7	2	207.5	−84.7	180.8	3.6	1R, 5
27	Dp	10	1	2	17.6	82.7	165.4	3.6	1N
28	Dp	11	0	0	21.4	79.9	356.8	2.4	1N
29	Dp	13	1	0	195.3	−80.7	136.3	3.4	1R
	Gb	0	0	4					
30	Gg	0	0	8					
31	Dp	14	0	0	11.1	76.9	119.8	3.7	1N
32	Dp	2	11	1	200.9	−82.8	91.1	4.4	1R

[a] D: dike; C: combined dike plus baked contact zone; G: granitic host rock; p: plagioclase-phyric; a: amphibole-phyric; l: low-temperature component; t: tonalite; g: granodiorite; b: baked/altered host rock in dike contact zone.

[b] e: end points; r: remagnetization circle end points; o: measured but omitted specimens.

[c] Precision parameter of Fisher (1953).

[d] Radius of cone of 95% confidence.

[e] 1: value used to calculate unit mean ChRM directions for a dike with reversed (R), normal (N) or mixed (M) polarity reported to its dominant polarity, for the dike's pyrrhotite component (P), or for a granite (G); 2: rejected as a deviant site mean ChRM direction; 3: K–Ar date of 51.9 ± 0.8 Ma; 4: K–Ar date of 45.0 ± 0.8 Ma; 5: K–Ar date of 50.1 ± 1.0 Ma.

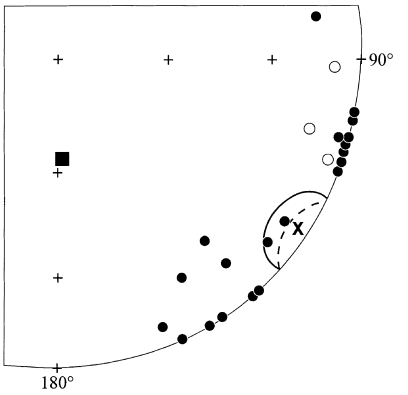


Fig. 3. Southeast quadrant of an equal-area stereogram showing the poles to the dikes' planes. Solid (open) symbols denote vectors in the lower (upper) hemisphere with the mean (X) and its cone of 95% confidence. The square symbol is the one pole excluded from the mean.

methods of orientation always agreed to $\leq 2^\circ$. From the cores, 10 to 18 specimens per site were sliced for measurement (Table 1), yielding a total collection of 407 specimens.

The dikes are intruded mostly into the 54 ± 2 Ma Fraser and 53 ± 2 Ma Summit Lake plutons (Fig. 2) and range from 0.5 to ~ 5 m wide. They vary in strike from roughly north–south to east–west with two dominant trends of east-southeast and south-southeast (Fig. 3). All but one dike have subvertical dips (Fig. 3). Excluding the one aberrant dike, the poles to the dike's planes have an average direction of declination $D = 125.5^\circ$ and inclination $I = 3.6^\circ$ with a radius for their cones of 95% confidence $\alpha_{95} = 9.2^\circ$. Had the host plutons been tilted since their emplacement, this mean and cone would not overlap onto the equatorial plane, given the wide range of strikes present. Thus this result suggests that the dikes were vertically intruded without subsequent tectonic tilt.

4. Petrography

The cores from each site were examined both in hand sample and in thin section. The mafic dikes are dominantly plagioclase-phyric diorites in composition (17 dikes), comprising 10 to 40% tabular plagioclase up to 6 mm long within a mixture of microcrystalline plagioclase and opaques. Biotite, opaques, amphibole and orthopyroxene are present in minor amounts ($<5\%$). The mafic minerals show minor alteration to chlorite, and calcic zones in the plagioclase are selectively sericitized. Five of the dikes are plagioclase-amphibole-phyric diorites, comprising up to 20% medium-grained (<3 mm) plagioclase and amphibole in a microcrystalline matrix of plagioclase, opaques and biotite. Minor alteration is found in most minerals. The three remaining dikes are medium-grained hornblende-phyric diorites, comprised of up to 40% medium- to coarse-grained (<6 mm) amphibole with fine- to medium-grained plagioclase in a microcrystalline matrix that also includes minor amounts of biotite, opaques and orthopyroxene. There is only minor alteration of the amphibole. The felsic plutons are fine- to medium-grained, massive, and homogeneous tonalites and granodiorites. Mafic minerals range from 5 to 40%, and include amphibole, pyroxene and biotite.

5. K–Ar age dating

Specimens from three dikes were analysed in the Geochronology Laboratory of the Department of Earth and Ocean Sciences at the University of British Columbia (Table 2). At site 26, a K–Ar whole rock age of 50.1 ± 1.0 Ma was obtained for a dike intruding the 59 ± 1 Ma Skagway pluton. A second whole rock age of 45.0 ± 0.8 Ma was determined for the dike at site 13 that intrudes the 53 ± 2 Ma Summit pluton. The third date, a K–Ar biotite analysis of 51.9 ± 0.8 Ma, was found for the dike at site 7 in the 54 ± 2 Ma Fraser pluton. Thus all dike ages are younger than their host rock ages, providing confidence in the validity of the determinations. The 51.9 Ma, 50.1 Ma and 45.0 Ma dikes are north-northwest-, northeast- and north-northeast-trending respectively. Note that only one dike, site 20, is mapped in the 48 Ma Clifton pluton. Assuming multiple ages of intrusion, averaging the three ages gives a mean value of 49.0 ± 2.9 Ma for the dikes for the purpose of selecting a reference paleomagnetic pole. If argon loss has occurred, then the two older ages probably

Table 2
K–Ar ages

Sample number[a]	Size (mesh)	K[b] (%)	^{40}Ar rad[c]		Age (2σ error)[d] (Ma)
			(mol/g $\times 10^{-10}$)	(% of total)	
SD95-07 Bi	−60+80	4.57	4.175	84.6	51.9±00.8
SD95-13 WR	−40+60	3.68	2.907	87.1	45.0± 0.8
SD95-26 WR	−40+60	1.87	1.648	83.3	50.1± 1.0

[a] SD95-07 at 59°43.29′ N, 135°02.07′ W is a coarse-grained diabase; SD95-13 at 59°41.81′ N, 135°05.10′ W is an aphyric green dacite; and, SD95-26 at 59°37.36′ N, 135°09.74′ W is plagioclase-pyroxene porphyritic andesite.

[b] Potassium was determined by atomic absorption using a Techtron AA4 spectophotometer on chloride solutions using a caesium buffer. K (%) is an average of at least two determinations.

[c] Argon was determined by isotope dilution using an AEI MS-10 mass spectrometer with Carey Model 10 vibrating reed electrometer, high purity ^{38}Ar spike, and conventional gas extraction and purification procedures as described by White et al. (1967).

[d] The errors reported, based on multiple analyses for K, are estimated two standard deviations as related to the calculated date. IUGS conventional decay constants (Steiger and Jäger, 1977) are $^{40}K_b = 4.962 \times 10^{-10}\,a^{-1}$, $^{40}K_e = 0.581 \times 10^{-10}\,a^{-1}$, and $^{40}K/K = 0.000\,116\,7$ atom ratio.

give the better age estimate of 51.0±0.9 Ma. In either event, the intrusion of the dikes occurred during the waning stages of emplacement and cooling of the host granitic plutons.

6. Paleomagnetism

6.1. Natural remanence

The natural remanent magnetization (NRM) of all specimens was measured on a Canadian Thin Films DRM 420 cryogenic magnetometer with a lower resolution limit for the remanence of $\sim 4 \times 10^{-6}$ A/m. The NRM measurements and all subsequent measurements were made inside a three-layer steel shielded room with an ambient magnetic field of about $\sim 1 \times 10^{-4}$ mT. The mafic dike specimens have site median NRM intensities ranging from 6×10^{-4} to 1.9×10^{0} A/m with an overall median $M = 2.3 \times 10^{-1}$ A/m and upper and lower quartile values (Q_3, Q_1) of 7.0×10^{-1} A/m and 2.4×10^{-2} A/m respectively. This wide range implies considerable variability in the composition of the dikes and, perhaps also, multiple ages of intrusion. Excluding specimens from the altered contact zones of the dikes, the granites give NRM intensity values of $M = 5.7 \times 10^{-2}$, $Q_3 = 1.7 \times 10^{-1}$ and $Q_1 = 2.3 \times 10^{-2}$ A/m.

6.2. Thermal step demagnetization

Initially, one specimen per site was thermally (Th) demagnetized in 12 steps up to 600°C using a Magnetic Measurements (MMTD-1) demagnetizer. The steps were concentrated between 260 and 320°C and between 500 and 600°C, the diagnostic demagnetization ranges of pyrrhotite and magnetite respectively (Fig. 4). The results proved straightforward. Following the initial removal of a viscous remanent magnetization (VRM), most mafic dike specimens gave relatively rapid intensity drops from ~525 to ~575°C, showing that magnetite is the main remanence carrier (Fig. 4a and b). Some dike specimens did show pyrrhotite signatures with distinct drops between 250 and 325°C also (Figs. 4c and 5f). The granitic specimens showed a more rapid rate of remanence decay in the VRM below 200°C and unblocking temperature spectra that indicate that titanomagnetite is the main carrier (Fig. 4d). In a very few cases, undemagnetized residuals at 600°C indicated the presence of minor haematite, which was confirmed by Th demagnetization up to 700°C. The characteristic remanent magnetization (ChRM) directions isolated in virtually all dike and granitic specimens were either of reversed polarity and directed steeply upwards to the south-southwest or normal polarity and antipodal (Fig. 5c and e).

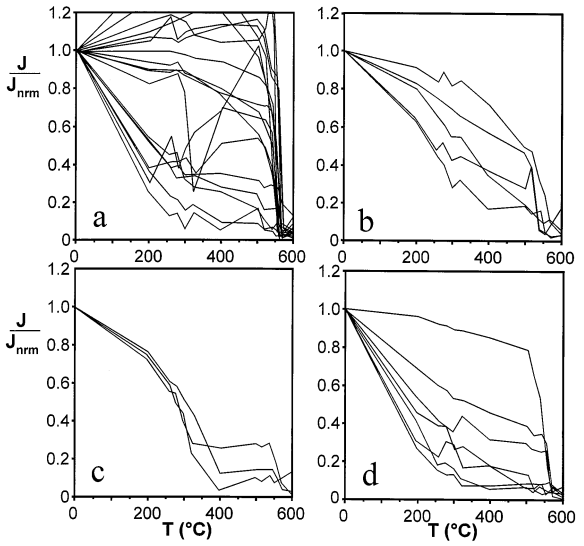


Fig. 4. Decay of remanence intensity on thermal demagnetization of specimens from: (a) reverse-polarity dikes; (b) normal-polarity dikes; (c) two-component (pyrrhotite and magnetite) dikes; (d) granitic sites. J/J_{NRM} is the ratio of measured intensity to initial (NRM) intensity. Temperature T is measured in degrees centigrade.

6.3. AF step demagnetization

Initially, AF demagnetization was done in 12 steps up to 130 mT on one specimen from each site using a Sapphire Instruments (SI-4) AF demagnetizer. For specimens with a normal ChRM direction, the VRM was reliably removed by $\sim$20 mT, whereas for those with a reversed ChRM direction, evident VRM was present up to $\sim$30 mT. Again, however, for nearly all of the specimens the isolated ChRM direction was either south-southwest and steeply upwards or antipodal (Fig. 5a and b).

6.4. Bulk demagnetization

Guided by the results of the initial demagnetization tests, most remaining specimens were AF and then Th demagnetized in about nine steps (Fig. 5d and g). The thermal steps again were focused on the 260 to 325°C pyrrhotite range and the 500 to 585°C magnetite range in particular. A minority of specimens (<30%) were subjected to either AF or Th demagnetization only, provided they gave well-defined ChRM directions.

6.5. Saturation remanence

Saturation isothermal remanent magnetization (SIRM) analysis was done on 20 specimens to examine the magnetic minerals. The test was done by pulse magnetizing them in 12 steps up to 900 mT in a direct current field using a Sapphire Instruments (SI-6) pulse magnetizer and then AF demagnetizing them in eight steps up to 140 mT. The remanence acquisition curves (Fig. 6a) are indicative of either single domain to multidomain magnetite or possibly single domain to pseudosingle domain pyrrhotite. The saturation remanence demagnetization curves (Fig. 6b) show that the ChRM is carried by single, pseudosingle and multidomain magnetite in the mafic dikes and granites, except in two granite specimens for which the curves flatten out notably beyond the single domain limit, thus indicating the presence of haematite also.

6.6. Statistical methods

Most specimens yielded well-defined ChRM end-point directions using the Kirschvink (1980) method. A number of specimens, mostly those with reversed directions, failed to reach stable end-points on demagnetization and so the remagnetization circle method of Halls (1978) was used to obtain an 'end-point' for them (Table 1). The site mean ChRM directions (Table 1) were calculated following Fisher (1953).

Only two of the dikes (sites #1 and 7) failed to yield a coherent cluster of ChRM directions; nearly all of the other dike sites have $\alpha_{95}<6^{\circ}$. Conversely two of the granite sites (#6 and 30) are incoherent, two more (#17 and 24) are loosely clustered with α_{95} values of 11.2° and 14.5°, and only one (#12) is reasonably well clustered with an α_{95} value of 6.9°.

Excluding contact zone specimens, 17 dikes have coherent reverse polarity, six have normal polarity, and two (#11 and 14) have mixed (i.e. both normal and reversed) polarity ChRM directions (Table 1). This polarity distribution is rea-

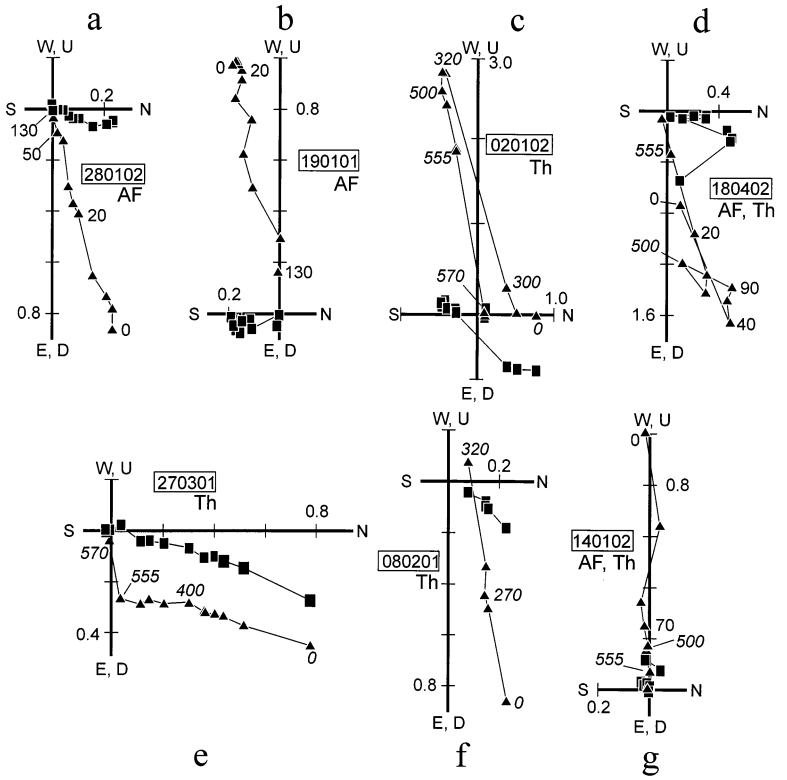


Fig. 5. Orthogonal vector component plots (Zijderveld, 1967) for example specimens on alternating field (AF) and/or thermal (Th) step demagnetization. The specimen numbers are in the box. The axes are: north (N), east (E), south (S), west (W), up (U) and down (D). The axial values are expressed as a ratio of the NRM intensity. Vector end-points are plotted as squares and triangles for the horizontal and vertical planes respectively. Some fields in millitesla (roman) or temperatures in degrees centigrade (italics) are labelled.

sonably consistent with emplacement during chrons 21R through 23 or from 47.9 to 52.4 Ma during which the Earth's magnetic field was reversed for ~63% of the time (Opdyke and Channell, 1996). For the coherent granite sites, one is normal, one is reversed and one (#12) has mixed polarity ChRM directions. On examination, the dike at site 5 was noted to have a mean direction that was deviant from the rest of the reversed site means (Fig. 7) and so it was excluded

Table 3
Unit mean remanence directions[a]

Group		N^* sites	Declination (deg)	Inclination (deg)	k	α_{95} (deg)
A	Reversed polarity dikes	16	199.4	−81.6	192	2.7
B	Mixed polarity dikes	2	191.5	−84.1	51	
C	Normal polarity dikes	6	9.2	76.5	318	3.8
D	All dikes	24	195.4	−80.6	168	2.3
E	Dikes' pyrrhotite component	2	6.1	75.5	77	
F	Granite plutons	3	3.6	80.8	50	17.5
G	Dikes plus granite	27	194.1	−80.6	146	2.3

[a] N^* is the number of sites averaged, as listed in footnote e (1) in Table 1. Other conventions as in Table 1.

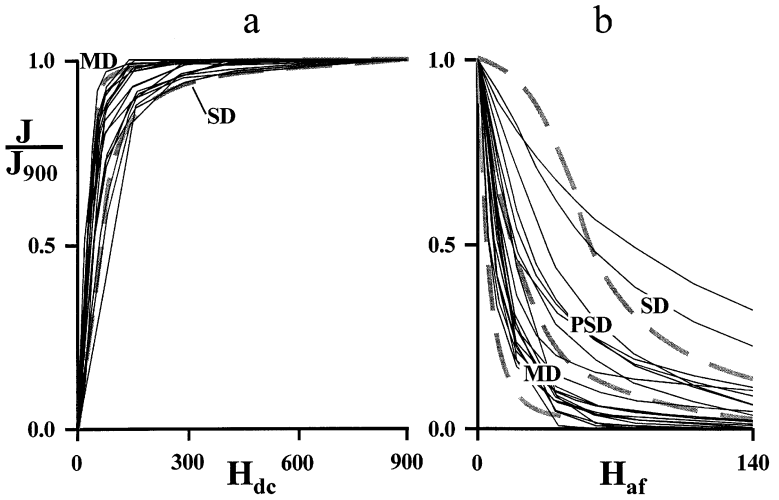


Fig. 6. SIRM curves showing: (a) acquisition of SIRM in a direct field (H_{dc}) up to 900 mT; (b) decay of SIRM on demagnetization in an AF field (H_{af}) up to 140 mT.

in calculating the unit mean directions (Table 3). It is also the dike with the aberrant attitude shown by the square symbol in Fig. 3. This suggests that the 'outcrop' may be a huge slump block in the mountain valley. The 24 accepted dike sites yield a mean ChRM direction of $D=195.4^\circ$, $I=-80.6^\circ$ ($\alpha_{95}=2.3^\circ$) after reporting the normal polarity sites to their antipodal direction, and this mean yields a paleopole position of 245.7° E, 76.3° N ($d_p=4.2^\circ$, $d_m=4.4^\circ$). This pole yields a translation estimate of $8\pm4^\circ$ S and a rotation estimate of $40\pm9^\circ$ clockwise (CW) when compared with the 50 Ma reference pole of Besse and Courtillot (1991) for the North American craton of 162° E, 77° N ($A_{95}=4.2^\circ$) that was derived by interpolation between their 42.8 Ma and 51.7 Ma pole averages (Fig. 10).

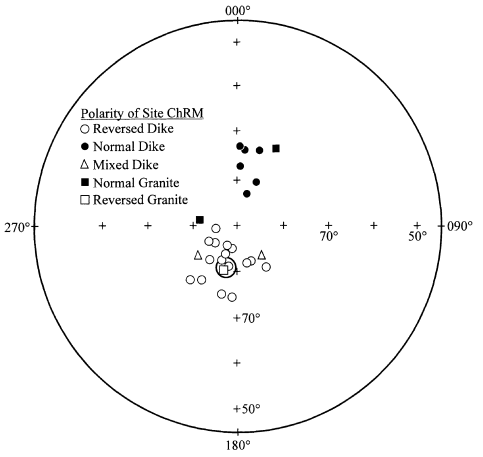


Fig. 7. Interior portion (>45°) of an equal-area stereogram showing site mean ChRM directions. Conventions as in Fig. 3.

6.7. Contact tests

Paleomagnetic contact tests were run at the five granite sites (Fig. 2). At these locations a dike site was collected with additional specimens from granite in its baked contact zone and a granite site was collected several dike-widths away from the dike. At sites 4, 5 and 6, a dike cuts a dike which in turn cuts the granite (Fig. 8a). Several tests were run because it was recognized that there was likely to be only a minor difference in time between emplacement of the host rocks and the dikes, so that the test would only work if a polarity reversal was captured. Three conclusive tests were obtained. The dike at site 11 gave a reversed direction, as did its baked zone, whereas the host granite at site 12 gave a coherent normal ChRM direction. The site 16 and 17 combination gave the same result. Similarly, the dike at site 4 has a normal ChRM direction; its contact zone gives a similar direction; and, the host dike at site 5 gives

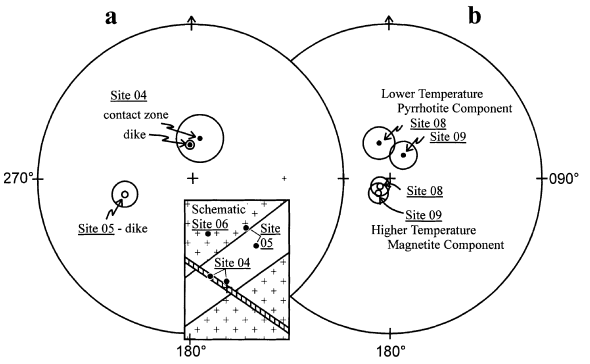


Fig. 8. Equal-area stereograms showing: (a) a paleomagnetic contact test for sites 4 and 5, where a mafic dike cuts a mafic dike in granodiorite host rock (see inset); (b) the site-mean high-temperature ChRM magnetite components for the dikes at sites 8 and 9, as well as for their lower-temperature pyrrhotite components.

a reverse polarity ChRM (Fig. 8a, Fig. 9), as does its baked contact zone in its granite host, whereas the adjacent granite at site 6 fails to retain a coherent remanence. These three positive paleomagnetic contact tests show clearly that the dikes and their host rocks carry a primary thermoremanent magnetization as their ChRM. The combination of sites 23 and 24 failed as a contact test because both dike and host rocks have the same reversed-polarity ChRM, and the combination of sites 29 and 30 also failed because the granite failed to retain a coherent remanence.

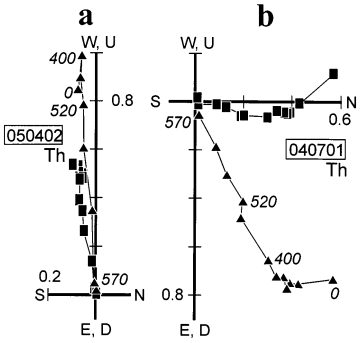


Fig. 9. Orthogonal vector component plots (Zijderveld, 1967) for specimens from: (a) the dike at site 5 with its typical reversed magnetization; and, (b) the baked zone of the normal-polarity dike at site 4 where it has been altered by the dike at site 5 (Fig. 8a). Conventions as in Fig. 5.

6.8. Pyrrhotite remanence

The dikes at sites 8 and 9 were found to have a measurable pyrrhotite remanence in most specimens on thermal demagnetization (Fig. 5f). Their direction is steeply down to the north, whereas the ChRM carried by the magnetite is about antiparallel in both instances (Fig. 8b). These results imply that the Earth's magnetic field switched polarity as the dikes cooled from $\sim$560 to $\sim$300°C, providing further evidence that the ChRM of the dikes is primary. Note also that both dikes required the extensive use of the remagnetization circle method to recover the site mean ChRM, suggesting that pyrrhotite may also have caused the need to use the remagnetization circle method for a few other sites (e.g. Fig. 5c).

7. Terrane tilt

On average, as noted earlier, the dikes have been emplaced vertically but have variable strikes, implying that the study area has not been tilted significantly or by more than $\sim$3°. Such a tilt, if post-intrusion, could alter the mean ChRM direction. If tilt occurred, then the rotation axis is likely to be NNW and orogen-parallel. Two further arguments suggest that tilt is minimal. First, the sampled transect is $\sim$35 km long astride the CPC so that any localized tilting is likely to be averaged out. Second, if those 12 dikes with vertical dips ($90\pm3°$) and accepted ChRM directions are compared with those 12 dikes with inclined dips ($<87°$), they give statistically identical mean ChRM directions, i.e. $D=199.5°$, $I=-79.4°$ ($\alpha_{95}=2.9°$) versus $D=191.2°$, $I=-81.6°$ ($\alpha_{95}=3.8°$) respectively. Had significant tilting occurred within the transect, then different directions would be expected.

8. Discussion

8.1. Post Mid Eocene translation

The cessation of northward motion for terranes underlying the southern part of the Intermontane

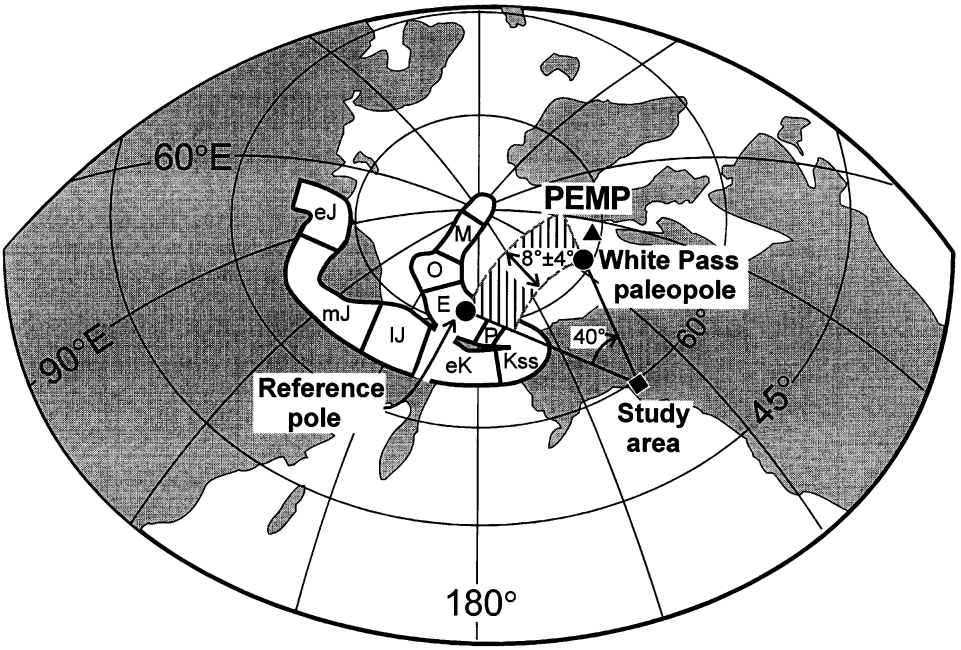


Fig. 10. The White Pass dikes' pole position relative to the 50 Ma reference pole on the apparent polar wander curve of Besse and Courtillot (1991). PEMP is the position of the present Earth's magnetic pole.

Belt relative to the craton is defined by four previous paleomagnetic studies (Table 4). Two are on rocks of the Quesnel Terrane. Fox and Beck (1985) reported paleomagnetic data for the Eocene Sanpoil lava flows at 85 sites, the underlying O'Brien Creek terrigenous sediments and tuffs at five sites and the overlying Klondike Mountain lahars, tuffs and lavas at 12 sites. From 18 K–Ar ages on plagioclase, biotite and hornblende separates reported by Pearson and Obradovich (1977), an average age of 51.5 ± 1.5 Ma is indicated for the unit mean ChRM direction (Table 4). Using a reference pole from Irving and Irving (1982), Fox and Beck (1985) concluded from their pole position of 304.6° E, 79.4° N ($d_p=5.3°$, $d_m=6.3°$) that their study area in north-central Washington State

Table 4
Comparison of Eocene tectonic motions

Unit		Age (Ma)	Translation (deg)	Rotation (deg CW)	Reference
1	Sanpoil volcanics	51.5 ± 1.5	3 ± 5 S	35 ± 7	Fox and Beck (1985); Pearson and Obradovich (1977)
2	Marron volcanics	52 ± 2	6 ± 7 N	9 ± 10	Bardoux and Irving (1989); Church (1978, 1981, 1982)
	Quesnel Terrane (1–2)	52 ± 2	2 ± 4 N	22 ± 13	
3	Dewdrop Flats volcanics	49.4 ± 2.4	7 ± 6 S	16 ± 2	Symons and Wellings (1989); Ewing (1981)
4	Ootsa Lake volcanics	49.6 ± 1.7	3 ± 7 N	25 ± 12	Vandall and Palmer (1990); Diakow and Koyanagi (1988)
	Stikine Terrane, (3–4)	50 ± 2	2 ± 6 S	20 ± 2	
	Intermontane Belt (1–4)	51 ± 1	0 ± 5	21 ± 10	
5	Eocene plutons	46 ± 2	7 ± 5 S	20 ± 11	Symons (1974, 1977)
6	White Pass dikes	51 ± 1	8 ± 4 S	40 ± 9	this study
	Eastern Coast Belt (5–6)	49 ± 2	8 ± 4 S	30 ± 10	
	Overall (1–6)	50 ± 2	3 ± 5 S	24 ± 10	

had not been translated northward relative to the North American craton but that it had been rotated ~25° CW between ~50 Ma and ~40 Ma. Compared with the 50 Ma reference pole for North America from the more recent Besse and Courtillot (1991) paper, this collection provides an estimate of $3\pm5^\circ$ of southward (S) translation and $35\pm7^\circ$ CW rotation for the collection area (Table 4).

Bardoux and Irving (1989) analysed the Eocene Marron volcanics and associated strata at 28 sites that also rest on rocks of the Quesnel Terrane in southern British Columbia. Five K–Ar biotite dates are available for these volcanics that give a mean age of 52 ± 2 Ma (Church, 1978, 1981, 1982). This collection gives a pole position of 197° E, 85° N ($A_{95}=10^\circ$) that provides an estimate of $6\pm7^\circ$ of northward (N) translation and $9\pm10^\circ$ CW rotation. Taken together these two studies from adjacent areas in the Quesnel Terrane indicate a nonsignificant translation of $2\pm4^\circ$ N and a significant rotation of $22\pm13^\circ$ CW.

The Dewdrop Flats volcanic rocks of the Eocene Kamloops Group rest on rocks of the southernmost Stikine Terrane in southern British Columbia. Fifteen K–Ar whole rock, biotite, plagioclase and sanidine age determinations from the group average 49.4 ± 2.4 Ma (Ewing, 1981). Based on paleomagnetic data from 282 specimens from 26 flows with reversals, conglomerate and fold tests providing evidence of the preservation of a primary remanence, Symons and Wellings (1989) determined a pole position of 221.6° E, 81.4° N ($d_p=11.0^\circ$, $d_m=12.3^\circ$). They noted that this pole was concordant with the Irving and Irving (1982) Eocene pole for North America and also concluded that northward translation had ended for the Stikine Terrane by this time. Using the updated Besse and Courtillot (1991) 50 Ma reference pole, the Dewdrop Flats result indicates $7\pm6^\circ$ S translation and $16\pm12^\circ$ CW rotation. In north-central British Columbia, the Middle Eocene Ootsa Lake Group also overlies rocks of the Stikine Terrane. Four K–Ar biotite dates give an average age of 49.6 ± 1.7 Ma (Diakow and Koyanagi, 1988). Vandall and Palmer (1990) obtained a pole position from 13 sites in subaerial lavas, ranging from basalt to rhyolite in composition, of 354.6° E, 88.0° N ($A_{95}=11.5^\circ$) having shown their remanence to be primary from fold and reversals tests. Using the reference pole from Bardoux and Irving (1989), they concluded that the Stikine Terrane had neither been translated northward nor rotated clockwise significantly since these lavas were extruded. Using the Besse and Courtillot (1991) reference pole, the calculated translation value is a nonsignificant $3\pm7^\circ$ N and the rotation value is a significant $25\pm12^\circ$ CW. Together, the two Stikine Terrane results give a paleolatitude discordance of $2\pm6^\circ$ S, implying no translation but a statistically significant clockwise rotation of $20\pm12^\circ$ CW.

All four paleomagnetic studies from terranes underlying the Intermontane Belt indicate a paleolatitude discordance of $0\pm5^\circ$, which strongly affirms that these terranes have not been translated relative to the North American craton since extrusion at about 51 Ma; however, they have had a significant rotational discordance of $21\pm10^\circ$ CW.

Two other paleomagnetic studies on Eocene rocks bear scrutiny. They are located on the eastern side of the Coast Belt and inboard of the Coast Shear Zone: one due east of Prince Rupert along the Skeena River, and the other along the fjords southwest of Kitimat in west-central British Columbia (Wanless et al., 1967; 1968, 1970; Symons, 1974, 1977). Combining data from both studies for the orogen-parallel NNW-trending 43 ± 5 Ma Kasiks diorite, 44 ± 4 Ma Alastair Lake granodiorite, 47 ± 7 Ma Ponder quartz diorite and 48 ± 2 Ma Quottoon diorite–quartz diorite yields 11 sites that give a mean direction of $D=358^\circ$, $I=76^\circ$, $\alpha_{95}=5^\circ$ (Symons, 1977). This direction is shown to predate deformation because the stable remanence of seven sites on the WNW-trending central limb of the Hawksbury warp is rotated ~50° in the correct sense to provide a regional paleomagnetic fold test. Also, antiparallel normal and reversely polarized sites are present. One caveat, however, is that no constraints on paleohorizontal are present to confirm that these NNW-trending plutons have not been tilted since emplacement; also, the demagnetization protocols were primitive by modern standards. Nevertheless, when compared with the cratonic reference pole, these plutons indicate a $7\pm5^\circ$ S translation and a $20\pm11^\circ$ CW rotation, values that are similar to

the White Pass result of $8\pm4^\circ$ S and a $40\pm9^\circ$ CW rotation (Fig. 10).

As summarized by Irving and Wynne (1992), all Miocene and younger pole positions derived from rock units in both the Intermontane and Coast belts give nearly concordant paleopoles with the craton, indicating neither rotation nor translation, thus supporting an Eocene end to significant translation.

The fact that the White Pass dikes and the Eocene plutons in west-central British Columbia both indicate marginally that southward translation occurred is not regarded as definitive. For the dikes, a tilt of a couple of degrees could easily be hidden in the error oval for the poles to the dikes' planes; and for the plutons, other than being sampled over a large area, which tends to average out tilts, there is no tilt control. Nevertheless, the White Pass dikes' result certainly argues against a northward translation. Thus we draw the conservative conclusion that there is no evidence for structurally significant northward translation inboard of the Coast Shear Zone–Denali Fault Zone in either the northern CPC or Intermontane terranes after about 51 Ma.

8.2. Intermontane Cretaceous–Paleocene data

Northward translation and clockwise rotation are well documented by paleomagnetism in mid Early to Late Cretaceous rocks with a primary magnetization from the terranes underlying the Intermontane Belt. As summarized by Harris et al. (1999) and using the Besse and Courtillot (1991) path for comparison, they include: from the Yukon, the 70 Ma Carmacks Group volcanics ($17\pm7^\circ$ N, $20\pm10^\circ$ CW; Marquis and Globerman, 1988; Johnston et al., 1996; Wynne et al., 1998), the 75 Ma Mount Lorne pluton ($11\pm4^\circ$ N, $57\pm11^\circ$ CW; Harris et al., 1999), the 109 Ma Mount McIntyre pluton ($14\pm5^\circ$ N, $79\pm16^\circ$ CW; Harris et al., 1996), and the $\sim$112 Ma Whitehorse batholith ($11\pm5^\circ$ N, $55\pm13^\circ$ CW; Harris et al., 1997); from central British Columbia, the 103 Ma Axelgod intrusions ($14\pm6^\circ$ N, $55\pm13^\circ$ CW; Monger and Irving, 1980); and, from southern British Columbia, the 104 Ma Spences Bridge volcanics ($14\pm7^\circ$ N, $66\pm13^\circ$ CW; Irving and Thorkelson, 1990; Irving et al., 1995). An end to northward translation by circa 51 Ma poses no problem for any of these results except for that from the Carmacks Group.

The Carmacks Group volcanics' result indicates an anomalously large northward translational displacement and low clockwise rotation. Assuming continuous northward displacement from the time of intrusion or extrusion to 51 Ma, the four mid Cretaceous plutons require minimum rates of motion between 1.8 ± 0.8 and 2.6 ± 1.1 cm/a, the 75 Ma Mount Lorne result gives a somewhat higher rate of 4.2 ± 1.4 cm/a, but the 70 Ma Carmacks Group result requires a much higher minimum rate of 10.4 ± 3.5 cm/a (Harris et al., 1999). In reporting the revised Carmacks result, Johnston et al. (1996) suggested that these volcanics originated atop of the Yellowstone hotspot plume at 70 Ma and were carried northward on basement fixed to the Kula plate. Recalculating the post-mid Cretaceous northward motion of the Pacific oceanic plates relative to the North American craton [see also Engebretson et al. (1985) and Kelly (1993)], Johnston et al. (1996) determined that the volcanics would reach their present location by 53 Ma with 100% coupling to the Kula plate. Suggesting that motion ended by 45 Ma, they reduced the required coupling to $\sim$70%. Applying a 51 Ma limit again pushes the coupling requirement up over 90%, a high value for a large terrane in a transpressional environment. Technical concerns have been raised concerning the Carmacks Group volcanics' result, such as the failure to average secular variation (Butler, 1990; Marquis et al., 1990; Harris et al., 1999). The basic problem is that either the Carmacks result is wrong in providing too large a translation estimate with minimal clockwise rotation, or all other Cretaceous results from the terranes underlying the Intermontane Belt are wrong. Put simply, if the mid Cretaceous results are right, and all four studies yield very similar estimates, then the terranes of the Intermontane Belt would have to have been translated southward from $\sim$110 to $\sim$70 Ma at a minimum rate of $\sim$1.8 cm/a before moving northward to satisfy the Carmacks result (Harris, 1998). This required southward transport conflicts with both the oce-

anic plate models of Engebretson et al. (1985) and Kelly (1993) and the northward tectonic motion required to satisfy the considerable Cretaceous paleomagnetic database for the Coast Belt and outboard terranes.

8.3. Post Mid Eocene rotation

A notable feature of the four paleomagnetic studies of Eocene rocks resting on terranes of the Intermontane Belt and of the two studies in the eastern side of the Coast Belt is that they all record amounts of rotation ranging from $9\pm10^\circ$ CW up to $40\pm 9^\circ$ CW with a mean of $24\pm10^\circ$ CW, and this rotation, on average, must postdate northward translation because they show no net poleward displacement. Over the years, numerous explanations have been offered to explain individual pole positions from Cordilleran rock units that give discordant CW-rotated positions. Such explanations include: an unrecognized primary dip, particularly in lava flows; the failure to measure sufficient sites to average out secular variation in the Earth's magnetic field (EMF), particularly in lavas; the failure to step demagnetize the specimens to a sufficient high AF intensity or temperature to remove all of the viscous remanent magnetization, resulting in a bias towards the present pole from a component of the present-day EMF on the reported direction; the unfortuitous capturing of an excursion in the EMF; or an unrecognized remagnetization caused by burial, adjacent plutons or hydrothermal fluid flow. Given that the database includes more than 200 sites over a large area that were mostly never deeply buried, all of these proffered explanations are certainly invalid for explaining the collective result. The possibility that the post-Triassic APWP for North America is wrong is discounted because of its match on tectonic construction to the paths of other continents (Besse and Courtillot, 1991; Van der Voo, 1993). Rejected also is the proposition that there was an omnipresent southwest-side down tilt about a horizontal or inclined northwest–southeast-trending axis that was about parallel throughout the orogen (Butler et al., 1989). Virtually all of the rotated Eocene and Cretaceous units listed above have either measured primary dips or other structural or geobarometric evidence to define or limit and correct for tilt.

Having rejected hypotheses based on geological or technological misadventures, on nonpresentation of the EMF and on tilts about horizontal or inclined axes, the remaining options must involve a rotation(s) about a vertical or near-vertical axis(es) after accretion.

The 'ball-bearing' hypothesis, first presented by Beck (1976, 1980), implies that the accreted terranes have been broken into a collage of small prismatic terranes, each a few to tens of kilometres across, that were rotated variably clockwise by dextral shear across the orogen. The model, as more recently stated (Beck, 1998), implies near-vertical faults in the rigid upper crust bounding each small prism, crushed-zones around the 'corners' of the prisms, and marked discontinuities between prisms in plutonic and structural trends. In general, mapping and deep crustal seismics do not appear to support this model for the terranes underlying the Intermontane Belt.

Irving et al. (1996) have suggested that the clockwise rotations record distributed dextral shear throughout the orogen. Such a hypothesis is, however, inconsistent with the massive character of the Eocene lavas and dikes and, in particular, of the mid Cretaceous plutons. To achieve the necessary $\sim60^\circ$ CW rotation of the remanence vector for the plutons, given the geometry of the orogen and location of the Cretaceous pole, would require extreme northwest–southeast elongation. The plutons were emplaced at relatively shallow depths and provide no evidence of such a fabric.

Recognizing difficulties with previous hypotheses to explain the clockwise rotations, leads us to suggest that sole-fault deflection tectonics may be responsible. Following Kelley (1993), a terrane located on an oceanic plate some hundreds of kilometres offshore of North America at ~110 Ma would be rotated $\sim40^\circ$ CW around an Euler pole relative to the North American craton prior to impact. Because North America was also moving northward, there would be little relative northward translation. Once the craton started overriding the Farallon–Kula plate boundary at ~83 Ma, the accreted terranes would move $\sim10^\circ$ N rapidly and be docked against the Canadian portion of the

North American margin by ~50 Ma. During the latter few Ma of the trip, the accreted terranes would also be driven up the ramp of the margin on sole faults. The LITHOPROBE deep crustal seismics profile of the southern Cordillera shows two critical features to support this hypothesis. The first is the presence of numerous subhorizontal sole faults throughout the crustal section (Cook, 1995), and the second is that the accreted terranes have been driven about ~300 km inland from the true edge of the craton (Monger, 1997). The plate models indicate that the impelling force on the terranes would be dextral transpression. As the terranes are driven mostly over and very slowly up the 'ramp' of the craton, they would experience a progressive increase in resistance in the south relative to the north as the crustal stack of slices is progressively thickened through thin-skinned tectonics. Thus additional variable clockwise rotation about vertical or near-vertical axes would be predictable, while still maintaining geologic coherence of structures and stratigraphic units over very large areas. Further, the model provides a mechanism for continued clockwise rotation after ~50 Ma for several million years without any northward translation. In the southern Intermontane Belt, the clockwise rotation could have been enhanced by post-accretion Eocene extension, which dies out in north British Columbia (Price, 1994).

Two critical tests for this model are in the offing. One is a paleomagnetic study of the ~54 Ma Flat Creek pluton near Whitehorse in the Stikine Terrane. If our model is correct, it should provide a similar result for the northern Intermontane Belt as the White Pass dikes of the CPC. The second test will come from the two LITHOPROBE deep-crustal seismic profiles being run across the northern Cordillera. This model predicts that major sole faults will be imaged and that the craton's edge will extend a few hundred kilometres southwest of the eastern edge of the accreted terranes.

9. Conclusions

The White Pass mafic dikes retain a primary dual-polarity thermoremanent magnetization that resides in magnetite and occasionally in pyrrhotite. Three K–Ar dates of 51.9 ± 0.8, 50.1 ± 1.0, and 45.0 ± 0.8 Ma constrain their emplacement into large felsic plutons of the CPC of Eocene age (mostly 59 to 48 Ma) that also carry a dual-polarity thermoremanent magnetization. Dike attitudes demonstrate that the study area has not been tilted since emplacement. Comparison of the 50 Ma reference pole for cratonic North America with the calculated pole position for the dikes of 245.7° E, 76.3° N ($d_p = 4.2°$, $d_m = 4.4°$) indicates a post-emplacement translation of $8 \pm 4°$ S and rotation of $40 \pm 9°$ CW. Overall, mid Eocene paleomagnetic data from this study, four studies in the Intermontane Belt, and two studies in the eastern Coast Belt indicate no significant northward or southward translation on average, but do indicate an average of $24 \pm 10°$ CW rotation. The latter motion is attributed primarily to sole-fault deflection tectonics of these accreted terranes after initial impact as they are driven up and over the cratonic margin.

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